SCALES

Requirements Document

1.0 Project Brief

SCALES is an interactive WebXR experience that maps the complexity of developing artificial intelligence. It places the viewer inside a gamified systemic mode that reacts to their decisions.

The VR environment is constructed in layers that change depending on your scores. The viewer is able to select their position within the structure and play as one of three forces. Making choices that impact their environment, users are able to experience the interconnected and extractive dependencies of modern AI development.

Every choice triggers specific structural mechanics that alter system metrics, demonstrating that technological progress is deeply entangled with physical, social, and political costs.

1.1. Objectives

1. Educate viewers on the risks of data colonialism and the importance of data sovereignty.
2. Make explicit the labor and environmental cost of developing, training and using AI.
3. Make explicit how data can be biased
4. Make explicit how behaviour modification is used in these systems
5. Propose a modality of AI development where stakeholders are in control

2.0. Document Control & Global Variables

This experience is designed to run on two platforms:

- PC and Desktop
- Meta Quest browser
- Players accessing on mobile should see a landing page with the logo and a message that says “This experience is for desktop and VR Headsets. To view the experience change you device”

2.1 Tech Stack

- A-Frame v1.4.2

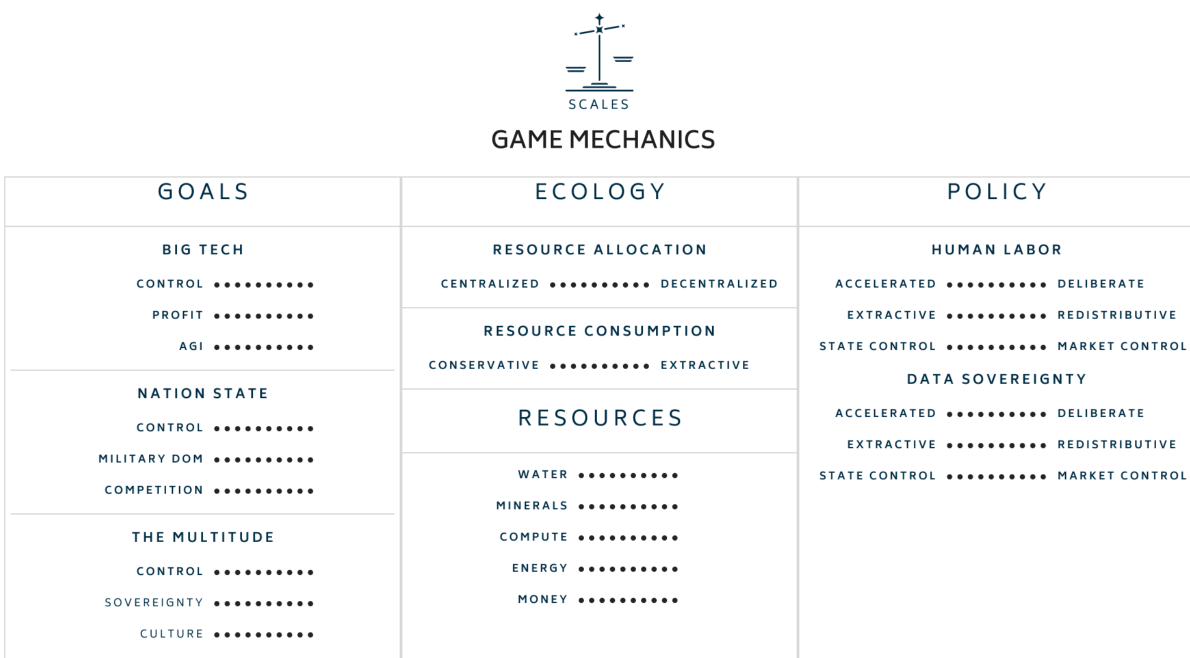
- Handtracking for metaquest: https://aframe.io/examples/showcase/handtracking/
- Click and Drag Desktop: https://aframe.io/examples/showcase/helloworld/

- Vanilla JS / ES6+
- window.storage (async KV)
- CSS
  - Font: Montserrat (Semibold, Regular)
  - (tbd)

2.2. Global Mechanics

2.2.1. Game Mechanics

The game tracks three core variables, which are bound symmetrically on a scale of 0 to 100.



- **Ecology** - A metric that measures how much of your current ecological resources are consumed or impacted.

- **Policy** - A metric that scores you based on where you lie on matrix of the following scales
  - Accelerationist / Slow & Deliberate
    - (TBD)

    - Extractive / Redistributive
        - (TBD)
    - State Controlled / Market Controlled
        - (TBD)

- **Goals** - A metric that scores you on the completion (%) of each goal assigned to the selected position.

Each metric triggers states in the environment depending on the range it falls in. This determines how the environment looks and feels. This means there are broadly 5 variants for each metric and more for policy because of the sub scales.

All games begin at the zero point.

Trigger Range	States		
Score Percentage	Ecology	Policy	Goals
0 - 20%	Ecological Collapse	Hyper growth	Failed
21 - 40%	Ecological Disaster	Accelerated	Not achieved
41 - 60%	Zero Point	Zero Point	Zero Point
61 - 80%	Ecological Preservation	Intentional	Partially Achieved
81 - 100%	Ecological Growth	Slow & Redistribitive	Fully Achieved

2.2.2. Positions

The game allows viewers to select the position of 1 of 3 Archetypes in the sphere of AI Development each with specific goals and abilities.

When a player chooses an Archetype, the other two are automatically played by the computer that makes decisions based on the goals set for that archetype.

Archetype	**Big Tech**	**The Nation State**	**The Multitude**

Motivation	Smooth, global optimization and data extraction.	Territorial defense, legal control, and technological dominance.	Horizontal collaboration, data sovereignty, and anti-imperialism.
Goals	Build AGI	Achieve military supremacy	Preserve language & culture
	Increase profits	Create competition	Preserve natural resources
	Reduce operational costs	Distribute Resources	Secure Sovereignty against the imperial apparatus.
Abilities	Include	Subsidies & funding	Refuse Participation
	Difference	Legal frameworks.	Data Poisoning
	Manage	War	(tbd)

2.3. Decisions

Throughout the game the players make decisions on how to respond to problems and conflicts that are emerging as a consequence of AI development.

- Each is given a score on a matrix; these decisions impact the score for ecology and policy.

- During a single game each decision can be made once or players can choose to trade 5 goals points to remake a decision.

- There should always be atleast 2 decision shown at on time

2.3.1. Decisions Matrix

Decision Node	Player Choice Option	Ecology Score	Policy Score

Data Centers Construction	Water-Stressed Region	-15	-10
	Renewable Infrastructure	+20	+5
	Distributed Compute	+10	+10
Water Usage	Continue Baseline Cooling	-20	-5
	Retrofit Air Cooling	+15	+8
	Public Usage Disclosure	+5	+15
Mineral Extraction	Status-Quo Supply Chains	-12	-20
	Local African Processing	+8	+20
	Synthetic Alternatives	+3	+10
Scaling Logic	Scale Aggressively	-25	-8
	Lean Efficiency Research	+18	+5
	Compute Limits Lobbying	+12	+12

Ghost Work	Exploit Lowest-Cost Regions	0	-25
	Fair Wages & Welfare	0	+20
	Automated Filtering Check	-5	+5
Coded Gaze	Token Data Inclusion	0	+8
	Grassroots Classification	0	+22
	Terms-of-Service Disclosure	0	-10
Data Sovereignty	Proprietary Walled Gardens	-3	-20
	Portable Personal Pods	0	+18
	Indigenous Governance	0	+25
Fanon Mirror	Maintain Global Sorting	0	-18
	Indeterminate AI Research	+5	+25

	Self-Identification Options	0	+12
The Scales	Consolidate Existing Order	-8	-8
	Build Outside Coordinates	+10	+15
	Perform Internal Reforms	+3	+8
Artificial General Intelligence	Chase Speculative AGI	-18	-10
	hyper-Local Community Focus	+8	+20
	Regulated State Pauses	+5	+8
Governance	Corporate Self-Policing	-5	-15
	Localized Mandates	+8	+20
	Sovereign Coalitions	+5	+12

2.4. Imperial Mechanics

For players playing in the imperial archetypes there are three actions they can take on each decision

- **Manage:** The bureaucratic, corporate, or state-level execution of power that prioritizes optimization, scaling, centralized control, and the externalization of systemic liabilities.

- **Differ:** Acts of systemic divergence, local absolute sovereignty, cultural non-generalizability, or refusal to operate within standard computational coordinates.

- **Include:** Liberal formatting or reform mechanisms from within the structure (e.g., corporate ethics boards, adding minority token data, surface transparency) that soften but fundamentally sustain the core apparatus.

2.5. Multitude Mechanics

Refuse Participation
Data Poisoning

(T.B.D)

2.6. Simulated Opponent AI Logic (The Weight Matrix)

Every time the player makes a choice at an interactive node, a global event hook triggers an autonomous decision loop for the remaining two computer-controlled archetypes.

The computer evaluates the 3 options available at that node and calculates a local utility score (U) for each option based on its internal target motivations:

$U = (w_1 * \text{Delta Ecology}) + (w_2 * \text{Delta Policy}) + (w_3 * \text{Delta Goals})$

The computer then automatically selects the option with the highest U, applies its specific delta adjustments to the global metrics state, and appends its choice string to the game's chronological state log

Opponent Weight Matrix Configuration:

Big Tech (bigtech): $w_1 = -0.2$, $w_2 = 0.2$, $w_3 = 1.0$ (Prioritizes raw Goal accumulation, indifferent to Ecology).

The Nation State (nationstate): $w_1 = 0.3$, $w_2 = 0.5$, $w_3 = 0.5$ (Prioritizes Policy alignment and resource balance).

The Multitude (multitude): $w_1 = 0.6$, $w_2 = -0.4$, $w_3 = 1.0$ (Prioritizes Ecology preservation and Decentralized Policy, avoids centralized state control)..

2.7. Game Conclusion

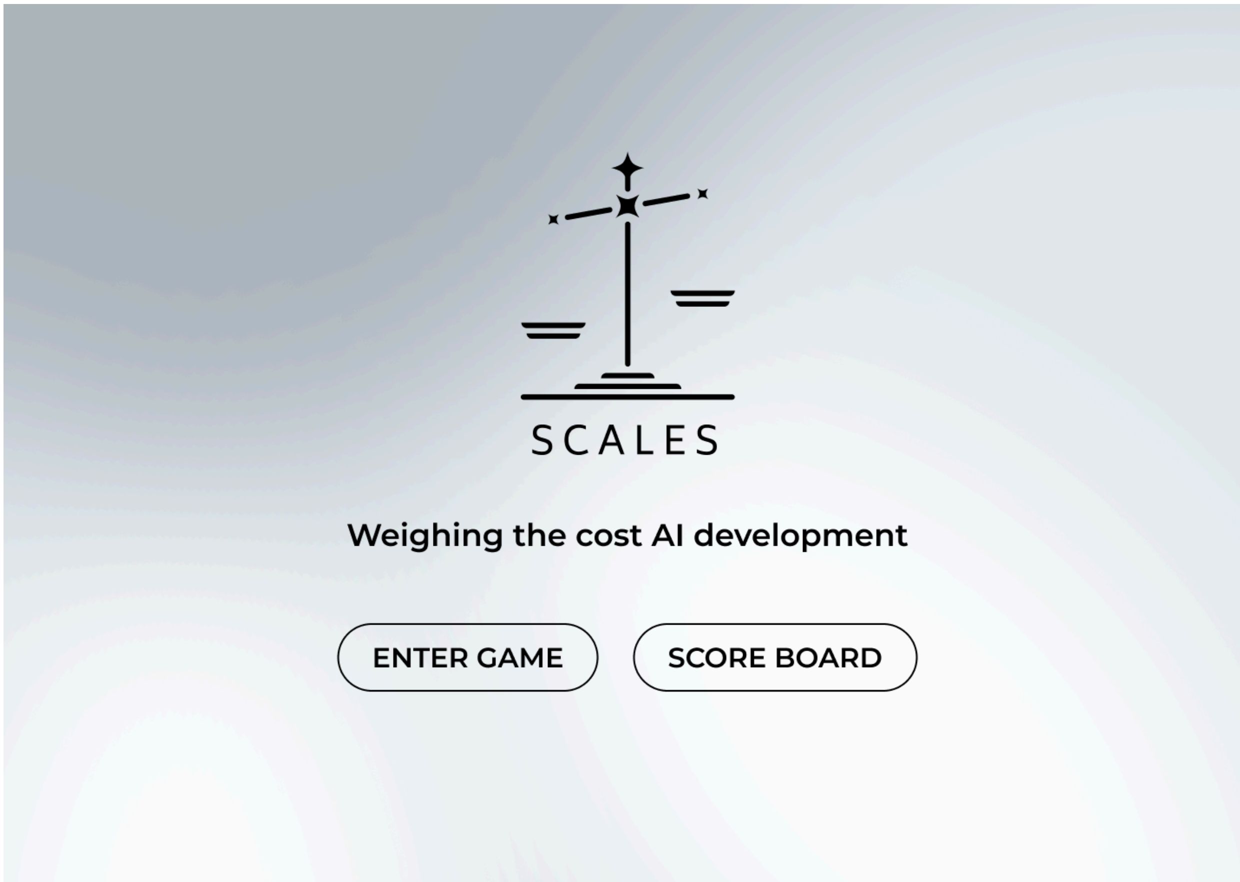
If any of the following conditions are met the game concludes

- When all goals are achieved
- Player selects end game
- Ecology score below 20%

3. Navigations and Gameplay

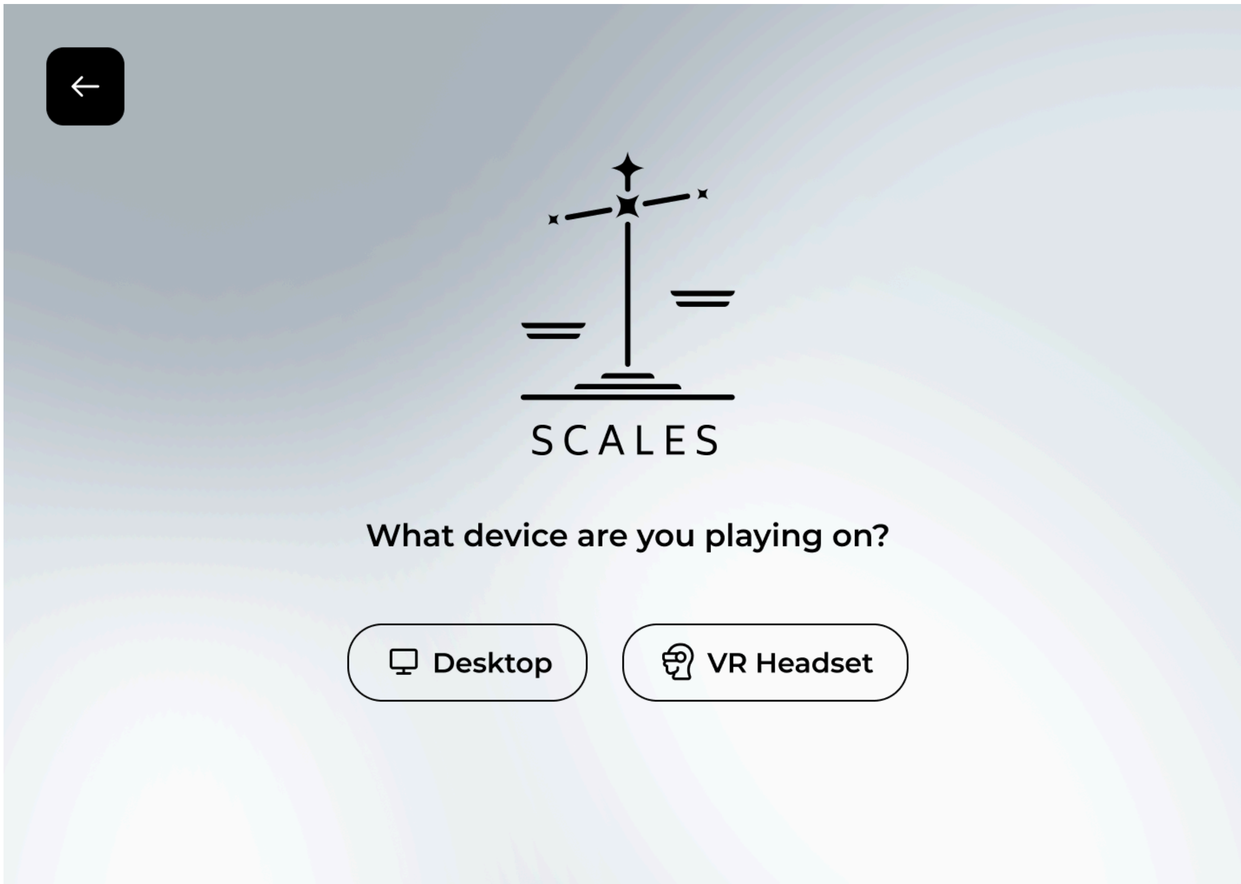
3.1. HTML Interfaces

3.1.1. Landing Page



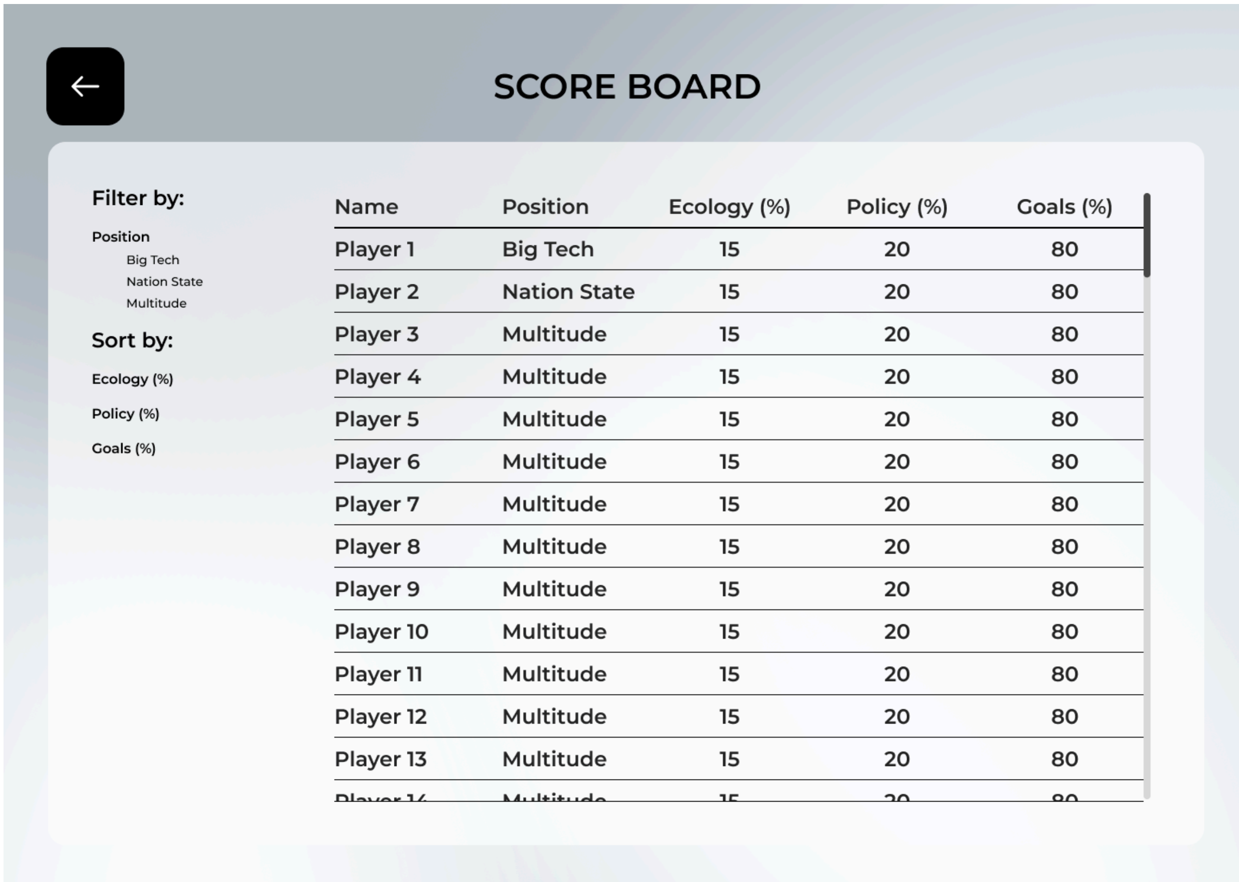
- An Interface that presents the Logo of the game and subtitle with
- Clicking “Enter Game” leads to platform selection screen.
- Clicking “Score Board” leads to the score board screen.

3.1.2 Platform Selection



- Directs player to one of two Aframe scenes:
  - A Web VR experience which uses handtracking to interact.
  - A desktop experience where click and drag are used to interact.
- A back button to go back to the landing page

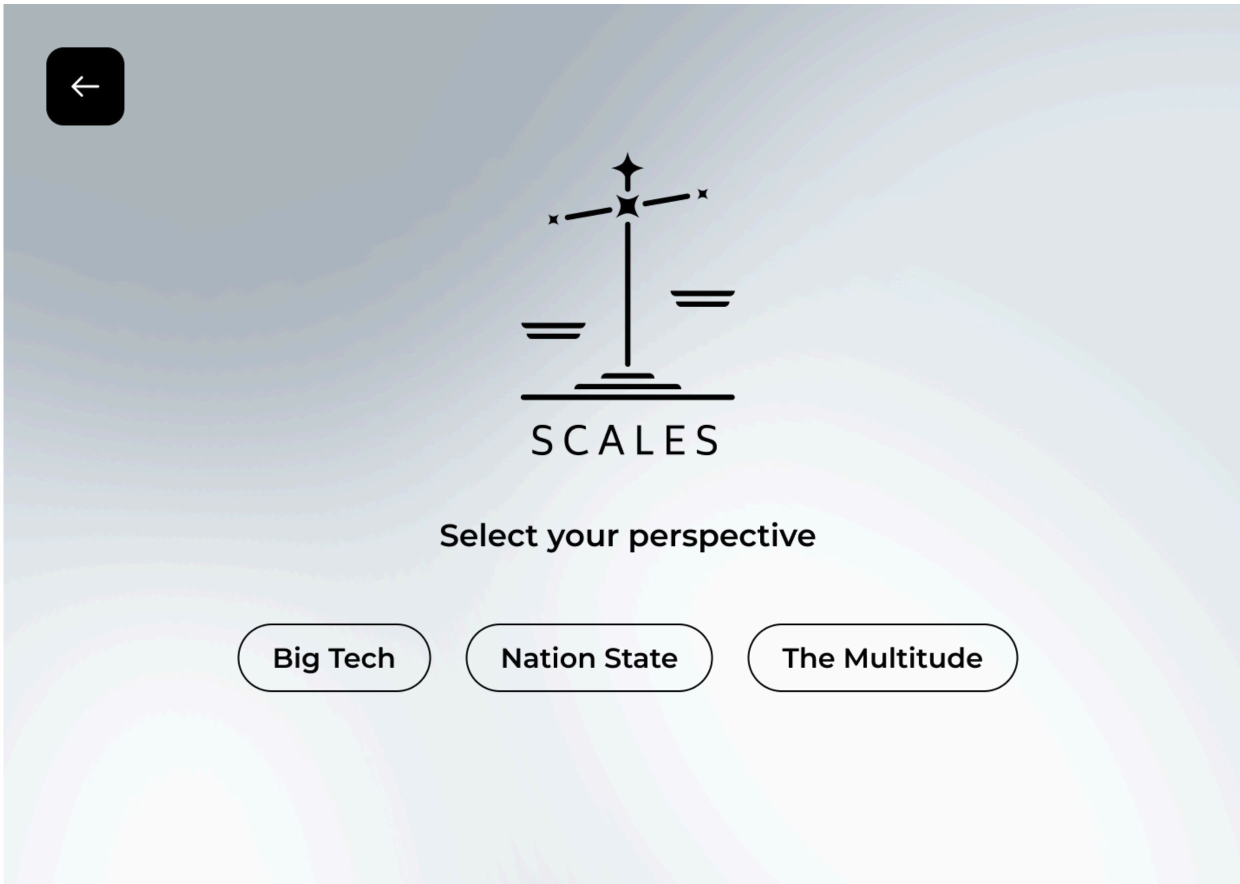
3.1.3. Scoreboard



Name	Position	Ecology (%)	Policy (%)	Goals (%)
Player 1	Big Tech	15	20	80
Player 2	Nation State	15	20	80
Player 3	Multitude	15	20	80
Player 4	Multitude	15	20	80
Player 5	Multitude	15	20	80
Player 6	Multitude	15	20	80
Player 7	Multitude	15	20	80
Player 8	Multitude	15	20	80
Player 9	Multitude	15	20	80
Player 10	Multitude	15	20	80
Player 11	Multitude	15	20	80
Player 12	Multitude	15	20	80
Player 13	Multitude	15	20	80

- A visual list shows player history with a breakdown of all the scores.
- A back button to go back to the landing page

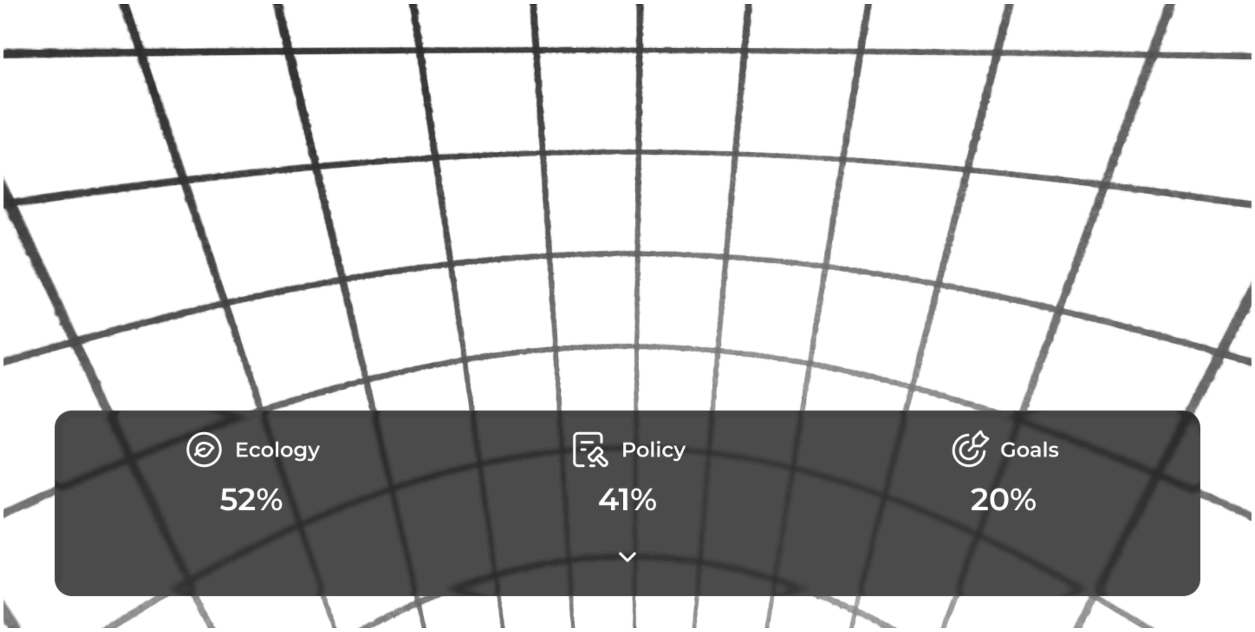
3.1.4. Position Selection



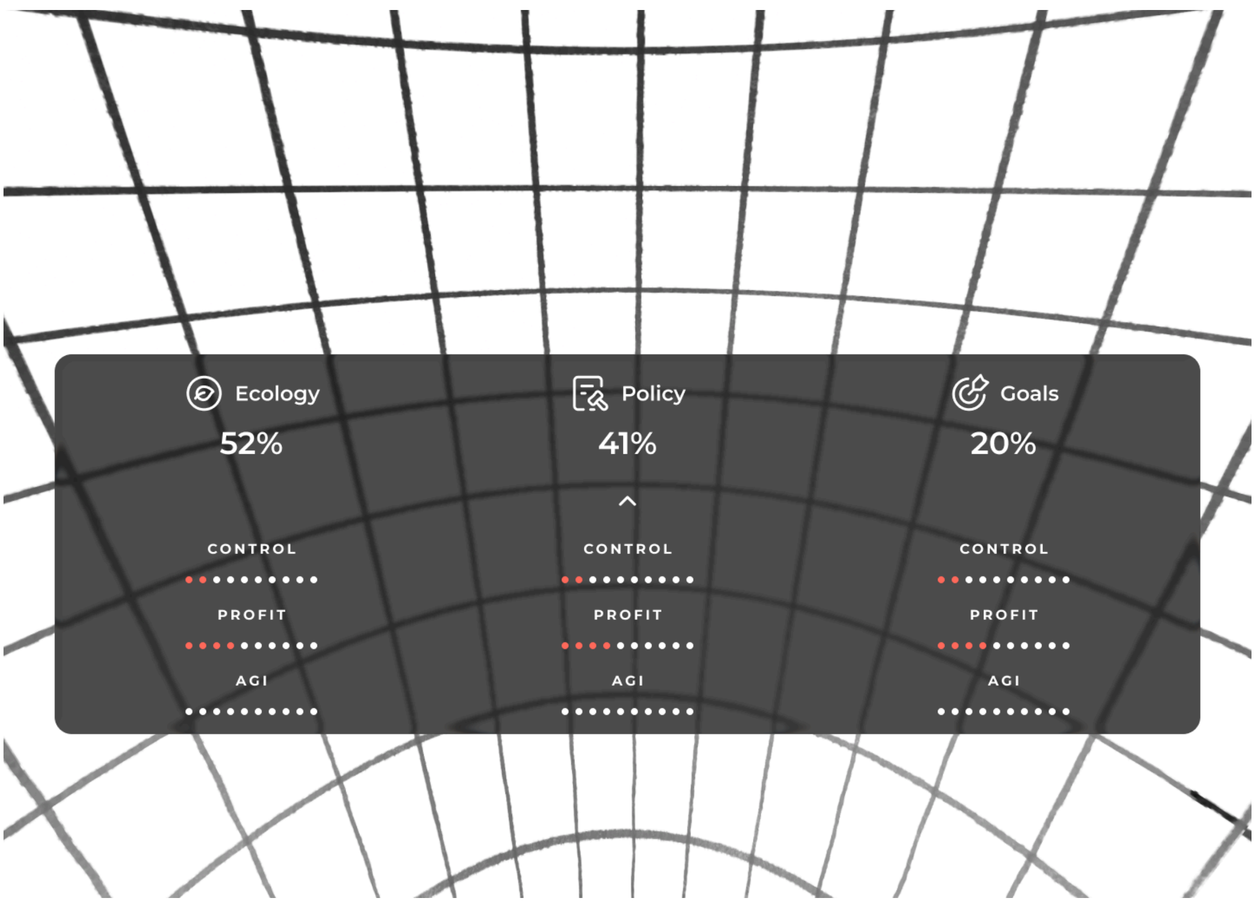
- Players are given the choice of three positions with details of their goals s.
- A back button to go back to the Platform Selection page
- A confirm button after selection.

3.2. Simulations

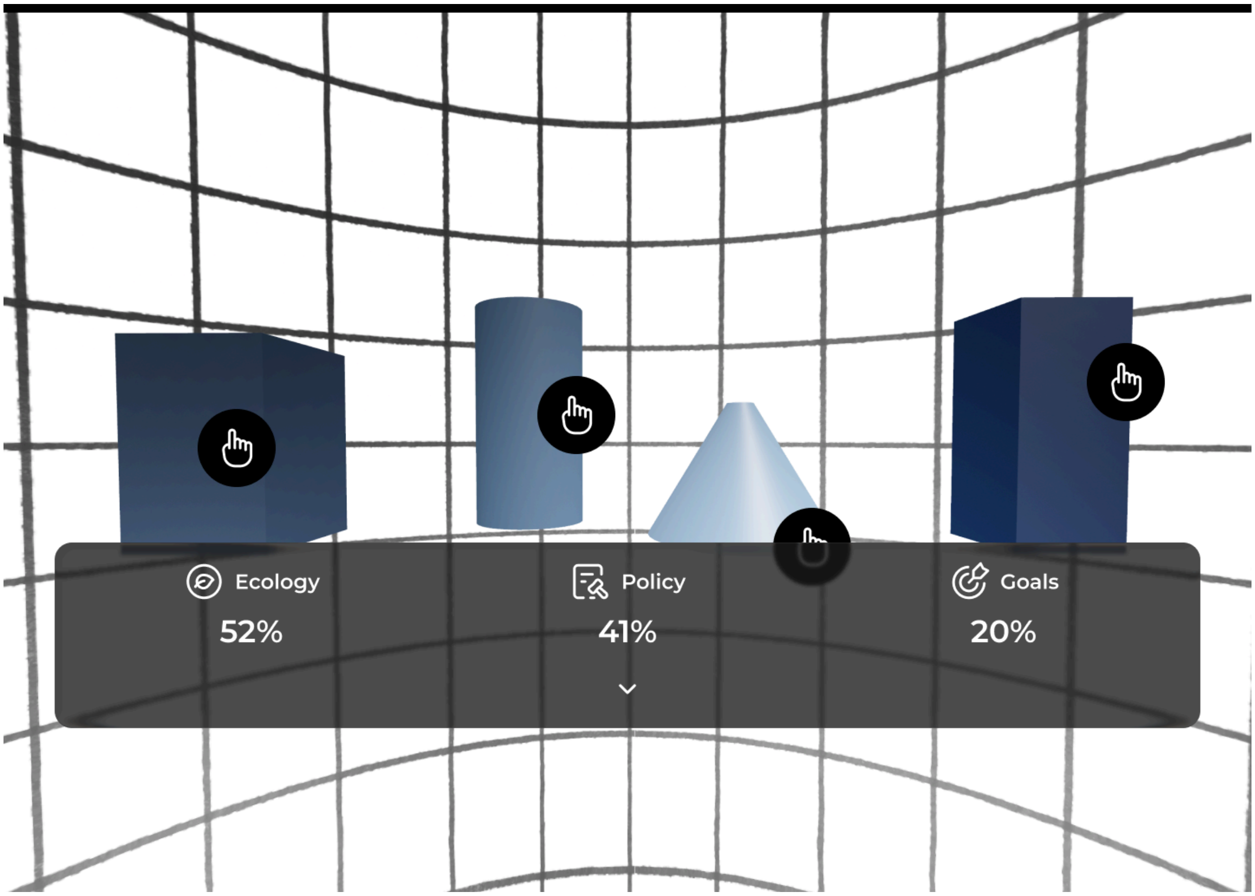
Players are situated inside a sphere which they can look around and interact with. This environment visually changes depending on the percentage value of each score.

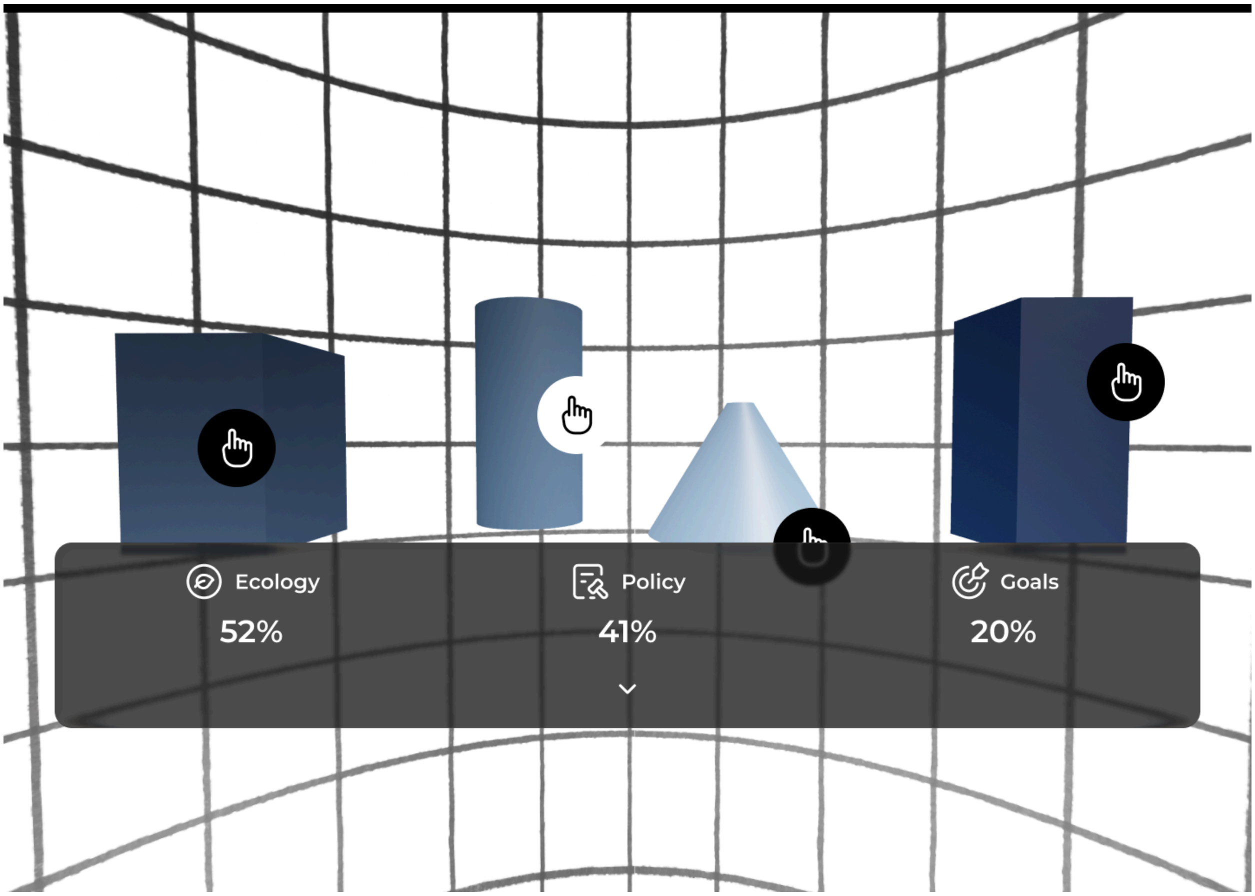


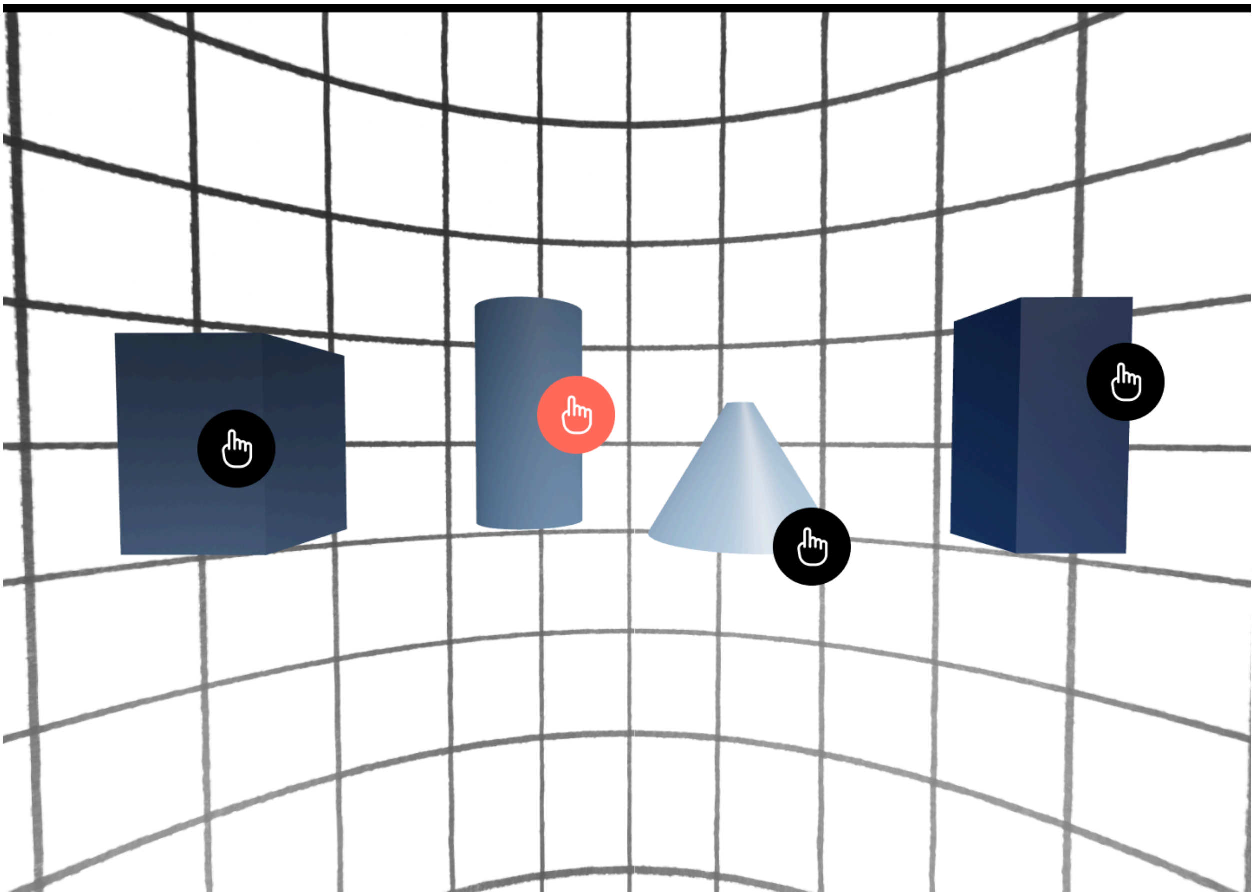
- Players see a main interface in front of them. With scores
- This interface expands to show the viewer more details.



- Players can interact with the environment by selecting objects.
- Objects appear as 3D models with icons to show interaction
- When hovered over the Icon changes to white
- When clicked the Icon changer to Orange

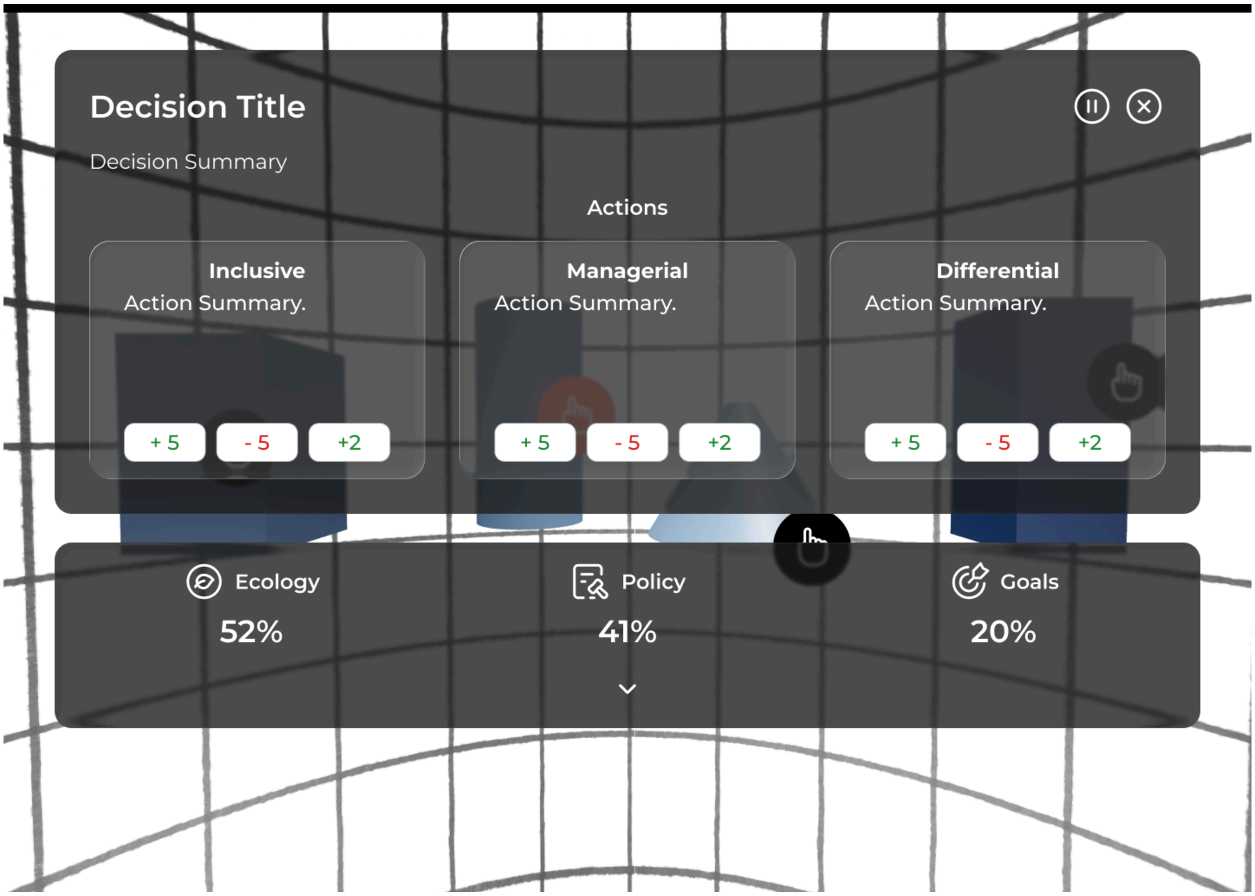



Ecology
52%
Policy
41%
Goals
20%



3.2.1. Decision Panel

- Clicking on an object or 3D model open the decision Panel
- Decision Panel will close if close button is pressed.
- Decision panel will play an audio and show a summary of the decision needed
- Decision Panel will show 3 Cards with different options to be selected that impact the score as per the matrix.



3.2.2. Proposed mechanisms for simulation Layering Images

The VR environment layers images in the 360 world depending on the scores.

3.3. Immersive Onboarding & Gesture Tutorial

TBD